

SUMMARY

PhD candidate specializing in **compiler optimization, parallel computing, linux-based network and real-time systems scheduling & security**. Built low-level infrastructure spanning real-time Linux kernel security modules, GPU scheduling frameworks, and compiler backends — with published work across systems security and HPC.

PUBLICATIONS

- **OpenMP-Q**: Quantum Task Offloading Extension for OpenMP — IWOMP 2025
- **T-Tex**: Timed Security in Multi-Threaded Real-Time Systems — ICCPS 2025
- **T-Pack**: Timed Network Security for Real-Time Systems — ISORC 2021

EDUCATION

PhD, Computer Science <i>Compiler Optimization, Embedded/Real-Time Systems, Parallel Systems</i>	North Carolina State University, Raleigh, NC <i>Aug 2020 – Present</i>
MS, Computer Science <i>Operating Systems, Linux Networking, Internet Protocol, Graph Theory</i>	North Carolina State University, Raleigh, NC <i>Aug 2018 – Apr 2020</i>
B.Tech., Computer Science <i>Parallel and Concurrent Systems, Cloud Computing, Agent Based Intelligent Systems</i>	Vellore Institute of Technology, India <i>Jul 2014 – Apr 2018</i>

WORK EXPERIENCE

AIM Intelligent Machines <i>Performance Optimization Engineer</i>	Monroe, WA <i>May 2024 – Aug 2024</i>
• Reduced CPU process latency by $\geq 80\%$ by identifying bottlenecks through targeted profiling (Perf, Nsight Compute, isolated execution analysis) • Implemented MPS-integrated GPU scheduling framework, cutting GPU latency step time by $\approx 9\%$ • Pinpointed critical CPU/GPU bottlenecks in a real-time robotics pipeline through systematic performance isolation	
Programming Systems, Uber <i>Compiler Optimization Engineer</i>	San Francisco, CA <i>May 2022 – Aug 2022</i>
• Implemented profile-guided code layout optimization (PGO) in the Go linker, improving cache utilization across large-scale multi-package benchmarks • Built a cross-package weighted call graph using C3 heuristics to generate optimal function ordering • Validated optimized layout against Google-scale Go benchmarks spanning multiple packages	
Defence Research and Development Organization <i>Cloud Infrastructure Engineer</i>	Agra, India <i>Dec 2017 – Jul 2018</i>
• Designed and deployed a cost-efficient infrastructure virtualization service using KVM hypervisors and CloudStack orchestration	

RESEARCH EXPERIENCE

Graduate Research Assistant

North Carolina State University, Raleigh, NC

Aug 2019 – Present

Research Advisor: Dr. Frank Mueller

OpenMP-Q: Quantum Task Offloading for HPC (IWOMP 2025):

- Designed and implemented an OpenMP language extension enabling seamless quantum task offloading within conventional HPC workflows, achieving up to **2.9× speedup** on representative quantum workloads
- Built a quantum-circuit library allowing users to define and execute circuits via POSIX pipes in Python
- Engineered for broad compatibility including reverse offloading and advanced scheduling features

T-TeX: Real-Time Security for OpenMP Applications (ICCPs 2025):

- Developed OpenMP code region identification using LLVM/Clang to detect and prevent timing-based delay attacks in real-time systems
- Designed a kernel-to-runtime communication interface to monitor context switches and secure timers, enabling worst-case execution time (WCET) evaluation
- Achieved **100% attack detection** for 60μs delay attacks with minimal runtime overhead

T-Pack: Network Security for Real-Time Systems (ISORC 2021):

- Built a packet modification layer using Linux kernel modules and socket buffers to analyze transmission timing and detect network intrusions
- Detected **95–100% of attacks** with a performance overhead of only $\approx 0.012\text{ms}$
- Extended to support all major network protocols with IPSec encryption compatibility

Advisor: Dr. Hung-Wei Tseng

In-Network Processing Analysis:

- Developed synthetic benchmarks using Apache Spark and Hadoop to surface network congestion bottlenecks
- Analyzed SDN performance for edge computing in 5G network scenarios

TEACHING EXPERIENCE

Graduate Teaching Assistant — Parallel Systems, NC State University

- Mentored graduate students on CUDA/NVIDIA GPU programming, MPI cluster programming, OpenMP/OpenACC multithreading, and system performance/power optimization

PROJECTS

Autonomous Vehicle Simulator (Carla):

- Designed a PID controller and CNN model to simulate autonomous driving; **100% of 10 concurrent vehicles** successfully followed traffic signs

Cloud Provider with DNS-as-a-Service:

- Built a virtual private cloud using Linux LXD containers with hierarchical DNS (bind9) and load balancing as default services

Peer-to-Peer File Sharing System:

- Implemented TCP/UDP file transfer with stop-and-wait reliability, achieving **95% network efficiency**

SKILLS

Languages: C, C++, Python, Go

Parallel & GPU Programming: OpenMP, MPI, CUDA, OpenACC

Compilers & Optimization: LLVM/Clang, DynamoRIO, Go Compiler, PGO, Profile-Guided Optimization

Systems Programming: Linux Kernel Modules, Network Stack, Process/GPU Scheduling, Signals, Timers

Distributed & Networked Systems: Apache Spark, Hadoop, SDN, IPSec, KVM, CloudStack